



A multicast router with unified buffers for network-on-chip: Low power and area optimization

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ABSTRACT

Network-on-Chip (NoC) has emerged as a viable communication framework for Multi-Processors System-on-Chip, thanks to its excellent communication efficiency. In real-world applications, besides unicast transmission, a substantial volume of multicast traffic also exists. The approach of handling multicast traffic via unicast-based mechanisms leads to numerous duplicate packets. Meanwhile, as chip fabrication processes continue to shrink, static power consumption due to leakage current becomes a significant challenge as transistor scaling continues, posing severe challenges to the thermal budget and energy efficiency of NoC design, necessitating urgent architectural optimizations. To address these issues, we propose a tree-based multicast routing algorithm. This algorithm ensures that both multicast packets and their copies reach destination routers through the shortest possible paths. For enhanced compatibility with the proposed routing algorithm, we also design a unified buffer multicast router architecture. This architecture not only simplifies the multicast routing algorithm but also reduces router power consumption and area overhead. Simulation results indicate that compared with traditional routers, the number of packets transmitted across the network decreases by 28%, while the area and power consumption are lowered by 40.6% and 38.6% respectively.

1. Introduction

Network-on-Chip (NoC) has become a promising communication paradigm for Multi-Processors System-on-Chip (MPSoC), attributed to its high communication efficiency, good scalability, and reliable performance [1]. Beyond one-to-one unicast traffic, NoC also handles one-to-many and one-to-all multicast data transmission. An increasing number of applications, such as high-performance computing (HPC), deep neural network (DNN) computation, and the cache coherence protocol (MESI), generate substantial multicast traffic within the network [2,3]. Utilizing traditional unicast routers to process multicast traffic results in a large number of duplicate packets, which quickly consume network bandwidth and cause network congestion. Meanwhile, with the growing number of cores in Chip Multi-Processors (CMP) and the continuous reduction of transistor sizes, researchers are confronting new challenges, particularly static power consumption due to leakage current becomes a significant challenge as transistor scaling continues. Power analysis at the 32 nm technology node reveals that static power consumption has tripled compared to designs based on the 65 nm technology [4]. Inefficient utilization of router resources and further transistor scaling may lead to even higher static power consumption in future designs.

In NoC architecture, data packets are transmitted through unicast and collective communication modes. Collective communication, including multicast and broadcast, is widely used in many high-performance practical applications, such as distributed memory synchronization and global data updates. Existing NoC research primarily focuses on one-to-one communication, with limited attention paid to multicast communication; this imbalance is partly due to the complexity of multicast routing, which involves packet replication and path optimization. Common multicast mechanisms include tree-based and path-based transmission approaches [5]. In tree-based multicast transmission, multicast packets are transmitted along shared paths as much as possible. They are then replicated at appropriate routers based on the router's location and the multicast routing algorithm. This mechanism offers a shorter average hop count but is prone to blocking, where congestion on one branch affects transmission on other branches. Furthermore, it requires addressing deadlock issues. In path-based multicast transmission, multicast packets select a single path that connects all destination routers, and replication only occurs after reaching a destination router. This mechanism is simpler to implement in hardware, minimizes the number of packets in the network, and

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is less susceptible to blocking, but it results in a longer average hop count. However, achieving fast transmission of both unicast and multicast packets requires adding numerous control components to routers, leading to increased NoC area and power consumption, which poses a new and severe challenge for NoC design.

Existing NoCs mainly adopt wormhole/virtual channel routers, which incorporate buffers at each input port to improve bandwidth efficiency during packet contention [6,7]. Nevertheless, buffer usage consumes significant chip area and static power [8,9], and maintaining efficient buffer utilization under varying workloads remains a challenge [10]. Common techniques for reducing static power consumption in NoCs primarily involve shutting down or sharing underutilized resources, and these methods can generally be categorized into two types: power gating and simplified microarchitecture.

Power gating is an effective method for saving static power by turning off idle circuit modules in the NoC [11]. However, the process of switching circuit modules on and off during power gating incurs non-negligible additional energy overhead. In cases of bursty traffic, power-gated circuit modules are frequently activated and deactivated, leading to extra delays. Recent studies have addressed these issues, but such solutions can only effectively save power at low injection rates [12–14]. Simplified microarchitectures achieve power and area savings by streamlining or eliminating certain router structures. Existing research mainly focuses on buffers [15–17], as buffers account for a relatively large portion of NoC area and power consumption. However, reducing the number and depth of buffers significantly limits network throughput.

To mitigate these problems, we propose a tree-based multicast routing algorithm aimed at improving the efficiency of multicast transmission in network communication. The NoC network is partitioned to split multicast packets, and a multicast address labeling strategy is employed to prevent excessive replication of multicast packets. The algorithm guarantees that all multicast packets and their copies follow the shortest paths to destination routers and is deadlock-free, eliminating the risk of network stagnation. Additionally, we design a router architecture that not only simplifies the process of determining packet transmission direction in the routing algorithm but also reduces router area and power consumption by merging redundant input buffers. The main contributions of this paper are as follows:

- (1) A multicast packet address partitioning method is proposed. Based on the source router's location, multiaddress multicast packets are divided into four multicast packets. This reduces the number of addresses per multicast packet and subsequently shortens the overall flit length.
- (2) A novel tree-based multicast transmission mechanism is introduced. It effectively reduces the number of multicast packet replicas in the network while ensuring each packet reaches its destination router with the same Manhattan distance hop count.
- (3) By merging input buffers, the complexity of input buffers and crossbar switches is significantly reduced, thereby lowering router area and power consumption.
- (4) Detailed simulation results demonstrate that the proposed multicast mechanism can efficiently transmit multicast packets while substantially reducing NoC area and power consumption.

The remainder of this paper is organized as follows: Section 2 introduces related work. Section 3 describes the multicast routing algorithm, and analyze the multicast algorithm's performance under different scenarios and discuss deadlock and livelock avoidance, respectively. Section 4 detail the router architecture and packet structure. Section 5 presents experimental results, and Section 6 concludes the paper.

2. Related work

Common multicast mechanisms mainly fall into three categories: unicast-based, path-based, and tree-based. Each category has its own advantages and limitations, and their applicability depends on specific application scenarios.

In [18], a unicast-based multicast mechanism is proposed. At the source node, multicast packets are split into as many unicast packets as there are destination addresses (for example, a multicast packet with 10 destinations becomes 10 unicast packets), which are then sent one by one to each destination router. Although this mechanism can be easily implemented without modifying traditional router architecture, as it reuses existing unicast routing logic, it may generate a large number of duplicate packets in the network. This quickly depletes network bandwidth, especially when the number of destinations is large, increases transmission delay due to increased contention, and reduces throughput (defined as the number of packets transmitted per unit time) since redundant packets occupy transmission resources.

To overcome the drawbacks of unicast-based multicast mechanisms, path-based and tree-based multicast mechanisms have been developed. In path-based multicast mechanisms, multicast packets are transmitted along a single path that connects all destination routers [19,20]; this path is precomputed before transmission begins, ensuring that all destinations are covered. In [21], a path-based connection-oriented multicast mechanism using wormhole switching is proposed. Wormhole switching splits packets into flits and transmits them sequentially, allowing partial packet transmission to reduce latency. Multicast packet transmission is divided into three stages: multicast establishment (where the path is computed and reserved), multicast communication (where packets are transmitted along the reserved path), and multicast release (where the path is freed for other uses). Packets are transmitted along the path established during the multicast establishment stage to reach each destination router, after which the path is released. Although this scheme is easy to implement, avoids deadlock (since the path is reserved and no cycles are formed), and minimizes the number of packets in the network, it only performs well when destination routers are geographically close. As the number of destination addresses in a multicast packet increases, the likelihood of packet bypass (where the path passes through non-destination routers), the length of the critical path (the longest segment of the path that determines transmission delay), and transmission delay all rise significantly.

To minimize packet transmission delay, tree-based multicast mechanisms have been proposed. In such mechanisms, multicast packets are transmitted along shared paths as much as possible, reducing the total number of transmission hops, and are replicated and rerouted when reaching tree branches [22,23]. The tree structure is dynamically built based on the location of destination routers, ensuring that each branch leads to a subset of destinations. In [24], a tree-based multicast mechanism uses point-to-point routing and recursive branching to transmit multicast packets within specific subtrees. Recursive branching means that each subtree is handled independently, simplifying the routing logic. Deadlock is avoided by limiting branching direction to downward, meaning that packets only branch to routers with higher coordinates in a predefined direction, preventing cycles. Compared to path-based multicast mechanisms, tree-based mechanisms are more complex, require higher hardware overhead (due to the need for branch decision and packet replication modules), and need to address deadlock issues. However, they result in fewer multicast packet replicas in the network and a lower average hop count for packets to reach destination routers, leading to reduced transmission delay. This is particularly advantageous in large-scale NoCs, where delay is a critical performance metric.

As transistor sizes in chips continue to shrink, static power consumption and chip area have become major challenges in chip design. Static power consumption, which is caused by leakage current, accounts for

Table 1
Comparison of different multicast routing mechanisms.

Feature	Unicast-based multicast	Path-based multicast	Tree-based multicast (Proposed)
Transmission strategy	Source router sends N separate copies, one for each destination.	A single packet traverses all destinations sequentially along a pre-computed path.	Packets travel along a common path and branch (replicate) at intermediate routers.
Packet replication	Occurs only at the source router.	No replication (single packet visits all).	Occurs at intermediate Routers (Branching points).
Latency (Hops)	Shortest path for each packet (Low), but high contention delay at source.	Longest path (High); latency accumulates linearly with destinations.	Minimal: Parallel transmission along shortest paths (similar to Unicast without source contention).
Hardware overhead	Low: Reuses existing unicast routing logic.	Medium: Requires logic to compute/store the path.	Optimized (Ours): Requires replication logic and deadlock handling.

an increasing proportion of total power consumption. In recent years, extensive research has been conducted to reduce chip static power consumption. Power gating is an effective method that saves power by turning off underutilized network components, such as idle input buffers or crossbar modules. However, the additional control elements required for power gating technology, such as sleep controllers and level shifters, introduce extra power consumption and area overhead. This overhead may offset some of the power savings, particularly in small-scale NoCs.

Simplifying the router microarchitecture of multicast routers is another effective way to achieve low overhead. In [15], a low overhead bufferless multicast router is proposed to support packet multicast transmission in bufferless NoCs. A 4×4 Banyan network is used instead of a serial crossbar to improve performance, as its regular structure allows parallel packet switching and reduces contention. Although the absence of buffers reduces area and power consumption compared to traditional buffered routers, it limits network throughput and causes significant packet delay due to multiple deflections at high injection rates. Deflection occurs when a packet cannot be transmitted to its desired output port and is redirected to an alternative port, increasing the number of hops. In [16], a shared buffer dataflow architecture router is proposed, allowing multiple input buffers to share data and buffer resources to improve utilization. This scheme separates the control and data parts of packets: input buffers store only the control part (which includes destination addresses and packet type), while a shared buffer holds the data part (the actual payload). Packets are reassembled into complete units before transmission, ensuring that the data part is correctly matched with the control part. Although this approach improves buffer utilization, it involves complex control logic to manage shared resources and requires additional delay and power consumption for packet splitting and assembly, which is undesirable for low latency applications.

To provide a clear distinction among the discussed approaches, we summarize the key characteristics of Unicast-based, Path-based, and Tree-based multicast mechanisms in Table 1. As shown in the table, while Unicast-based methods suffer from high traffic load and Path-based methods experience high latency, the Tree-based mechanism (adopted in this work) offers an effective balance by minimizing both packet replication and hop count.

3. Multicast routing algorithm

3.1. Quadrant division

To ensure fast packet transmission in the network, unicast packets use the traditional XY routing algorithm, while multicast packets employ the proposed tree-based multicast routing algorithm. Based on the source router's position, we logically partition the entire network into four quadrants. Each multicast packet is permitted to be replicated and transmitted solely within one quadrant. As shown in Fig. 1, the entire mesh network is logically divided into four quadrants centered on the source router (Router 12). Except for the source router, every other router in the network belongs to exactly one quadrant. For routers located on network edges or vertices, some quadrants may not contain any routers under this division method, but this does not impact the functionality of the designed multicast routing algorithm.

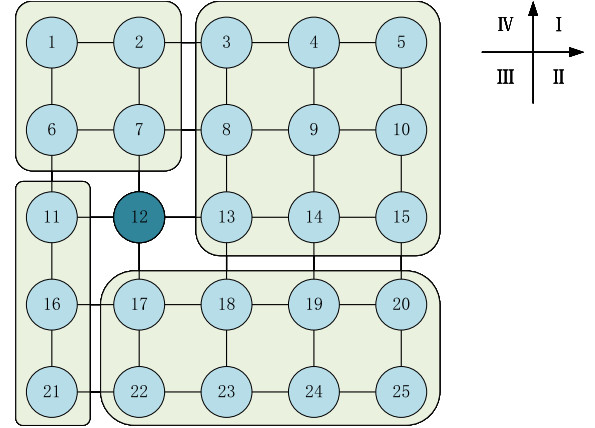


Fig. 1. Diagram of quadrant division.

3.2. Multicast address division and marking

Long multiaddress packets are split into multiple multicast packets, each containing up to four addresses. This division is based on router coordinates within the mesh network's quadrants. If the number of destination addresses is large, multiple four address multicast packets can exist within a single quadrant, but each multicast address can only appear once. As shown in Fig. 2(a), Router 12 acts as the source router, sending packets to destination routers 3, 7, 14, 16, 20, and 22. Excessively long packets require more flits, increasing both transmission time and the risk of contention. To mitigate this, the source router splits the 6-address multicast packet into four 2-address sub-packets based on the destination quadrants, as shown in Fig. 2(b). These sub-packets retain identical payloads and source addresses. For example, destinations 3 and 7 are in the north quadrant, so they form one subpacket; destinations 14 and 20 are in the east quadrant, forming another subpacket, and so on.

Meanwhile, to track whether a multicast packet or its copy has been sent to each destination address, a one bit address Flag is added before each destination address, initially set to 1. A Flag value of 1 indicates that the corresponding destination has not yet received the packet, while a value of 0 indicates that the destination has received it. The source router then sends each of the four multicast packets to the corresponding quadrant using the proposed routing algorithm, and as packets are replicated and transmitted, the Flag bits are updated to reflect the transmission status.

3.3. Routing algorithm

After copying multicast packets into their respective quadrants, the source router first sends each packet to the router in the quadrant that is closest to it. Upon reaching this nearest router, if there are still destination addresses with an unset Flag (indicating unsent packets), the current router is treated as a new source node. Specifically, when a multicast packet is split, the Flag vector is updated via a bitwise

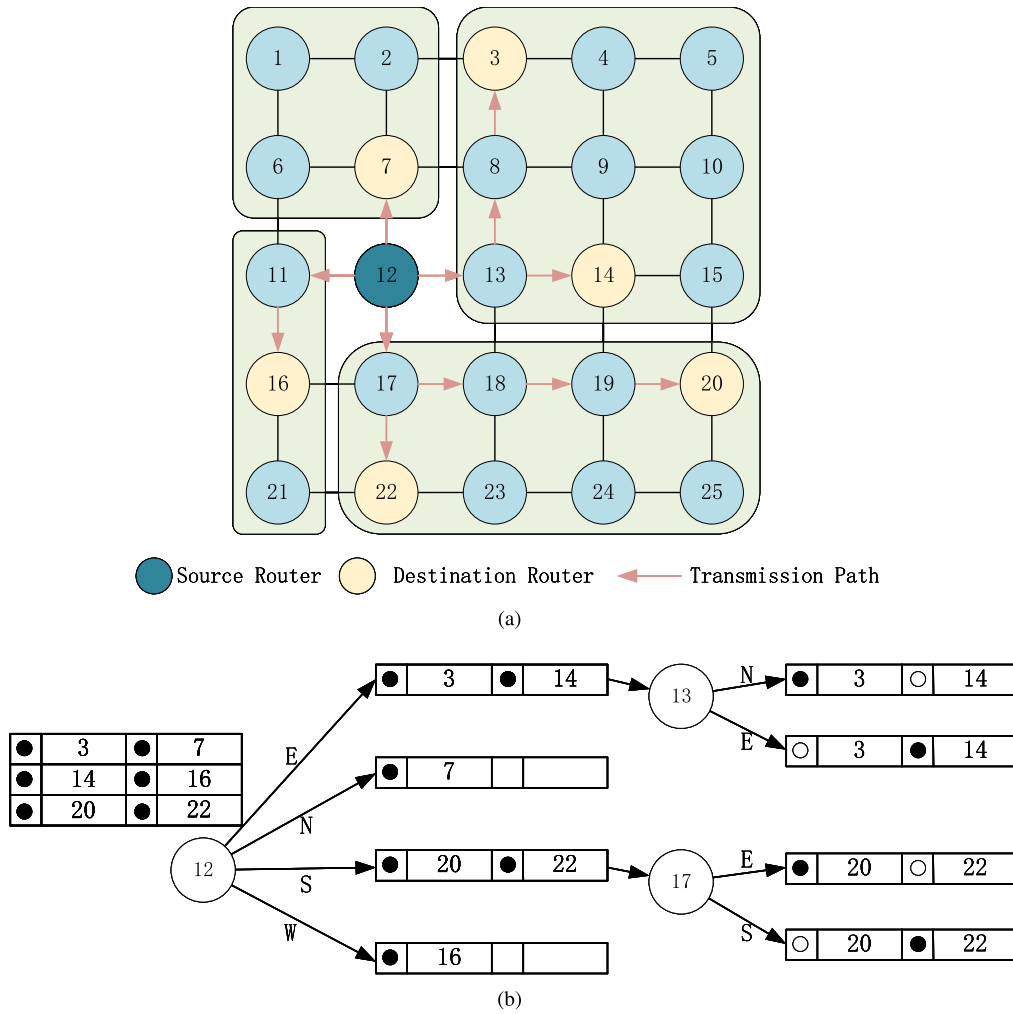


Fig. 2. Examples of routing algorithms: (a) packet transmission path, (b) multicast packet address division and marking.

AND operation with the destination mask of the target quadrant, ensuring that the new subpacket only carries the addresses relevant to its direction. Specifically, the Flag modification process is implemented using a bitwise AND operation. The router maintains a predefined Direction_Mask for each quadrant or output port. When a multicast packet needs to be replicated to a specific direction (e.g., East), the router performs $\text{New_Flag} = \text{Original_Flag} \& \text{Direction_Mask_East}$. This operation ensures that the replicated packet only retains the flag bits corresponding to the destinations reachable through that direction, while other bits are masked to 0. This mechanism prevents redundant routing and ensures that the packet stops forwarding once all its target destinations have been reached (i.e., when Flag becomes all-zeros).

To ensure multicast packets follow the shortest path during each transmission and prevent excessive replication, the following packet copying rules are established:

- (1) When a multicast packet enters the current router from the X direction (W input and E input ports), check if there are any destination routers in the Y direction within the router's quadrant. If so, copy the packet and modify its destination address Flag; otherwise, continue routing.
- (2) When a multicast packet enters the current router from the Y direction (N input and S input ports), check if there are any destination routers in the X direction within the router's quadrant. If so, copy the packet and modify its destination address Flag; otherwise, continue routing.

According to the proposed multicast routing algorithm (as shown in Fig. 2(a)): In the first quadrant, the multicast packet is first sent to the nearest router (Router 14). During transmission, when passing through Router 13 (via the W input port), it detects that Router 3 (in the Y direction) is a destination router. The multicast packet is then copied, its address Flag is modified, and it is output through the E and N output ports of Router 13. Upon reaching Router 14, if there are still unreached multicast addresses, the above steps are repeated with the closest remaining destination address as the target. In the second quadrant, the multicast packet is first sent to Router 22. When passing through Router 17, the packet is replicated, its address Flag is modified, and it is then sent to Routers 20 and 22 respectively.

The detailed algorithm is described in Algorithm 1. The input includes the packet header (containing Type, Flag[4], Dst1, Dst2, Dst3, Dst4), XY_cur (coordinates of the current router), and bf (indicating whether the current packet is buffered in the X direction or Y direction buffer). The output includes STBE, STBN, STBW, STBS, STBL (crossbar selection signals), Modify Flag (transmitted to the address marking unit, representing the Flag of the replicated multicast packet), and Cur Flag[4] (the address marking of the current multicast packet that needs modification before transmission).

3.4. Situation analysis

The proposed tree-based multicast routing algorithm logically divides the entire mesh network into four quadrants. The source router splits multicast packets into multiple 4 address subpackets based on

Algorithm 1 Routing Algorithm

Input: XY_cur , bf , $Type$, $Flag[4]$, $Dst1$, $Dst2$, $Dst3$, $Dst4$
Output: STB_E , STB_N , STB_W , STB_S , $Modify_Flag[4]$, $Cur_Flag[4]$

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1 if  $Type == 01$  then
2    $XY$  routing algorithm to generate the nexthop  $STB$  signal for
    $Dst$ ;
3 end
4 else if  $Type == 10$  then
5   Select the address closest to the current router with a  $Flag$  of 1,
   noted as  $Dstj$ ;
6   if  $STB_L == 1$  then
7      $Flag[j] = 0$ ;
8      $Cur\_Flag[4] = Flag[4]$ ;
9   end
10  for  $i \leftarrow 0$  to 4 do
11    if  $bf$  is the  $X$  direction buffer and  $Dsti$  is the same with
       $XY\_cur$   $Y$  direction then
12      if  $X$  coordinate of  $Dsti$  is greater than  $X$  coordinate of
         $XY\_cur$  then
13         $STB_E \leftarrow 1$ ;
14         $Modify\_Flag[i] \leftarrow 1$ ;
15      end
16    else
17       $STB_W \leftarrow 1$ ;
18       $Modify\_Flag[i] \leftarrow 1$ ;
19    end
20  end
21  else if  $bf$  is the  $Y$  direction buffer and  $Dsti$  is the same with
       $XY\_cur$   $X$  direction then
22    if  $Y$  coordinate of  $Dsti$  is greater than  $Y$  coordinate of
       $XY\_cur$  then
23       $STB_N \leftarrow 1$ ;
24       $Modify\_Flag[i] \leftarrow 1$ ;
25    end
26  else
27     $STB_S \leftarrow 1$ ;
28     $Modify\_Flag[i] \leftarrow 1$ ;
29  end
30 end

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destination router locations, with each subpacket responsible for transmitting to all destination routers in its corresponding quadrant. This partitioning ensures that each subpacket operates within a limited range, reducing the impact of multicast traffic on the entire network.

Fig. 3(a) illustrates the best case scenario of the algorithm. In this case, each of the four quadrants contains at least one destination router. For example, the north quadrant includes Router 3, the east quadrant includes Router 14, the south quadrant includes Router 22, and the west quadrant includes Router 7. Packets are sent to each quadrant simultaneously, and within each quadrant, the initial branch router is close to the source, minimizing the initial transmission hop count. Packets reach all destinations along the shortest path simultaneously, with no redundant paths or duplicate packets. This results in effective reduction of transmission delay, as each destination receives the packet in the minimum number of hops, and network bandwidth is used efficiently, since no redundant packets are generated. In this scenario, the average packet delay is approximately equal to the Manhattan distance from the source to the farthest initial branch router, which is significantly lower than path-based mechanisms that require a single path covering all destinations. Fig. 3(b) shows the worst case scenario.

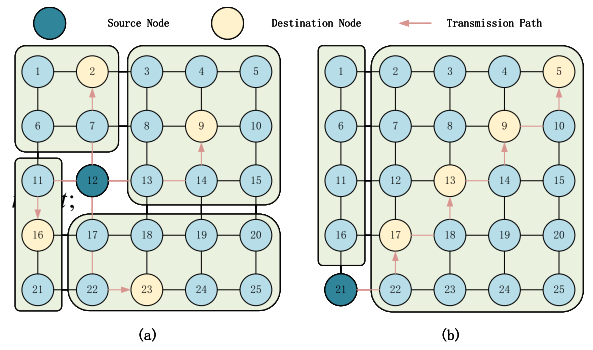


Fig. 3. Situation analysis: (a) Best case, (b) Worst case.

In this case, the source node is located at a corner of the mesh network. For example, the top left corner (Router 1), and all destination routers (such as Routers 14, 15, 20, 22) are concentrated in one quadrant (the southeast quadrant). During packet transmission, there are no routers capable of efficiently replicating packets along the way, this means that the initial branch router (Router 22) is far from the source, and the packet must travel the entire distance to the initial branch router before any replication occurs. For example, the packet from Router 1 to Router 22 must travel 8 hops (6 south and 2 east) before reaching Router 22, where it is replicated to Router 20 and other destinations in the quadrant. However, even in this case, the hop count for all destination routers remains the same as that of the Manhattan path. For example, Router 20 is 7 hops from the source (5 south and 2 east), and the packet reaches it via Router 22 in 8 (to Router 22) + 1 (from Router 22 to Router 20) = 9 hops. The algorithm ensures that during transmission to the initial branch router, any intermediate routers that can replicate to nearby destinations will do so. In reality, even in the worst case, if Router 17 is on the path to Router 22, and Router 20 is a destination, Router 17 will replicate the packet to Router 20 when the packet passes through, so the hop count to Router 20 is the Manhattan distance (7 hops), not the sum of hops to the initial branch router plus replication. Thus, the worst case hop count still equals the Manhattan distance, ensuring that even in unfavorable conditions, the algorithm does not increase delay beyond the minimum possible.

3.5. Deadlock and livelock avoidance

In tree-based multicast routing mechanisms, packet replication and forwarding can lead to deadlocks, making deadlock avoidance an essential component. Common deadlocks in NoCs are mainly classified as route-level and network-level deadlocks.

Route-level deadlocks typically occur within a router. When multiple packets need to be forwarded from the same output port (which can only handle one packet at a time), contention and mutual waiting between packets may arise, resulting in a route-level deadlock. To avoid this, we incorporate Virtual Channels (VCs) in the input buffer, similar to traditional routers.

Network-level deadlocks usually occur across the entire network. Forwarding paths between multiple routers form a loop, with each router waiting for the next router to release resources. This can prevent packets from being forwarded properly, leading to a network-level deadlock. Although the designed router architecture supports multicast packet transmission, the proposed routing algorithm selects the nearest router in each quadrant as the destination router. Packet transmission within each quadrant still uses the turn restricted XY routing algorithm, and replicated packets do not require turns to reach their destinations. Thus, the proposed routing algorithm is network-level deadlock-free.

Livelock in NoCs refers to packets continuously deflecting in the network and never reaching their destination nodes. Unicast packets use the XY routing algorithm, which is livelock-free. For multicast packets:

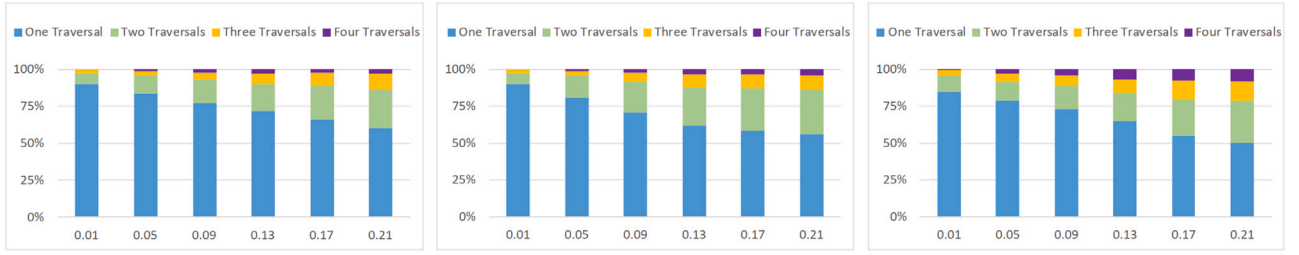


Fig. 4. The number of input buffers simultaneously accessed at different packet injection rates.

- (1) After transmission to each partition, they are first sent to the destination node with the shortest hop count (using the XY routing algorithm, which avoids livelock).
- (2) After reaching a destination router, the next transmission targets the closest remaining destination router in the partition (again using the XY routing algorithm).
- (3) Packets replicated during transmission are aligned with the destination router in either the X or Y direction of the current router and do not require deflection to reach their destinations.

Thus, the proposed algorithm is also livelock-free.

3.6. Input buffer access rate

Traditional NoC routers are equipped with an input buffer for each of the four input directions (east, south, west, north). To increase network throughput and prevent deadlocks, input buffers contain multiple Virtual Channels (VCs). Packets pass through the buffer and crossbar to reach the output port. The crossbar allows multiple packets to enter the traversal phase simultaneously. Simultaneous access is defined as the event where request flits arrive at multiple input ports during the same clock cycle. We monitored the input ports cycle-by-cycle to calculate the probability of concurrency.

We observed input buffer access patterns in NoCs with multicast traffic. Fig. 4 shows the simultaneous access of input buffers in different router directions at varying injection rates. Fig. 4(a), (b), and (c) represent scenarios where multicast packets account for 0%, 30%, and 60% of the total data flow, respectively. At low injection rates, nearly 80% of router input buffer accesses occur in a single direction. Simultaneous access in two directions is rare, and access in three or four directions is nearly non-existent (close to 0%). As the injection rate increases, the number of packets in the network grows, but access is still dominated by one or two directions. Even in worse cases, simultaneous access in three or four directions occurs in less than 20% of instances. Clearly, even with multicast traffic, the four input buffers of routers are underutilized.

To fully utilize network resources and ensure compatibility with the proposed multicast routing mechanism, we propose a shared buffer router architecture. By merging X direction and Y direction buffers, this architecture not only improves resource utilization but also aligns better with the proposed routing algorithm. This is because the algorithm only copies packets when:

- (1) A packet is transmitted in the X direction with a destination router in the Y direction.
- (2) A packet is transmitted in the Y direction with a destination router in the X direction.

Merging X and Y input buffers eliminates the need for the router to determine packet transmission direction, as X and Y direction packets are stored in separate merged buffers. Buffer merging reduces the number of crossbar input ports from five to three, simplifying the crossbar structure.

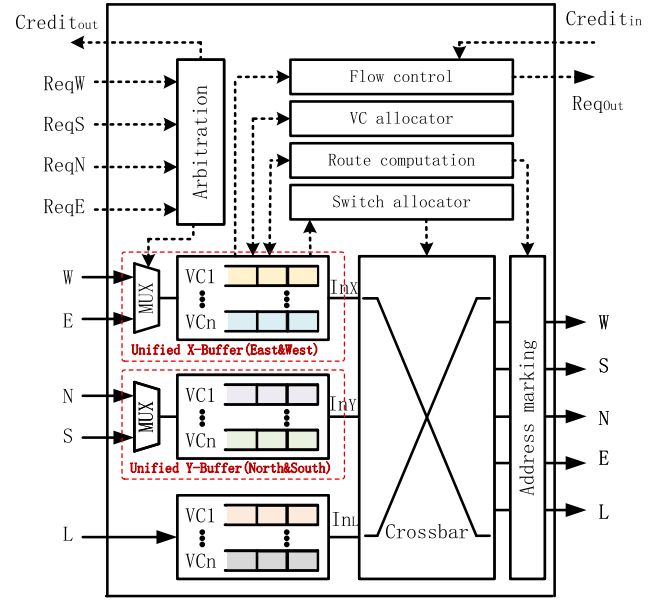


Fig. 5. Multicast router architecture.

4. Router architecture

The router architecture (shown in Fig. 5) consists of an input buffer, an arbitration unit, a flow control unit, a virtual channel allocator, a route computation unit, a crossbar allocator, an address marking unit, and a multiplexer. As noted earlier, the probability of simultaneous input buffer access in multiple directions is low. Thus, there is no need to assign a separate buffer to each input port. To prevent contention due to insufficient buffers, we only merge buffers from two input directions. For ease of calculation in the multicast routing algorithm, X direction (W and E) input buffers are merged, and Y direction (N and S) input buffers are merged. The merged W/E buffer stores X direction transmitted packets, and the merged N/S buffer stores Y direction transmitted packets. This eliminates the need for the multicast routing algorithm to determine packet transmission direction.

The arbitration unit receives input request signals from upstream routers and generates multiplexer selector signals and Credit_out signals. The flow control unit sends Req_out signals to downstream routers and receives Credit_out signals from downstream routers. Together, the arbitration and flow control units form the router's complete traffic control mechanism. The virtual channel allocator and crossbar allocator have similar functionality to those in traditional routers. In addition to generating crossbar selection signals, the route computation unit also generates new multicast packet addresses to send to the address marking unit. To prevent infinite replication of multicast packets in the network, an address marking unit is included to track destination addresses that have received multicast packets.

Table 2
Experimental parameters.

Parameter	Value
Network size	$8 \times 8, 12 \times 12$ 2Dmesh
Router algorithm	Unicast, multicast
Channel width	128 bits
Buffers per port	16(Banyan is 0)
VC	4(Banyan is 0)
Multicast packets	4 multicast addresses
Multicast traffic	4, 8, 12 or 16 multicast addresses

Table 3
Accelerators parameters.

Parameter	Settings
Topology	6×6 Mesh(400 $\times$ 400 $\times$ 100 ANN) 8×8 Mesh(Lenet-5) 10×10 Mesh(AlexNet) 16×16 Mesh(VGG-16)
Routing	X-Y Routing(Base)
Buffers per port	12(3 virtual channels)
PE performance	86.4GOPS
Frequency of router	1 GHz
Memory bandwidth	2 GB/s
Channel width	128 bits

- (3) Banyan [15]: A bufferless router architecture that supports multicast. It eliminates input buffers entirely to save area and power, instead using a reconfigurable Banyan crossbar to deflect packets in case of contention, if a packet cannot be sent to its desired output port, it is redirected to an alternative port. Banyan uses a simplified multicast routing algorithm that replicates packets at the crossbar, but the lack of buffers leads to higher delay at high traffic loads.

We compared packet transmission delay and network throughput of UBMR and the other three schemes under random traffic with varying percentages of multicast and unicast packets. Simulation configurations are shown in Table 2.

In recent years, NoC-based DNN accelerators have been widely studied [5,27,28]. Since the output of neurons in the previous layer of a deep neural network is sent as input to all neurons in the next layer, NoC-based DNN accelerators generate substantial multicast traffic. A suitable multicast mechanism can significantly improve accelerator performance. We used each of the above multicast mechanisms as the multicast packet processing scheme for the accelerator and compared accelerator classification delay and the number of transmitted packets under different schemes. Since traditional Noxim does not support accelerator simulation, we modified Noxim following the methods described in [29–31]. Accelerator simulation configurations are shown in Table 3.

5.2. Average packet delay

Average packet delay is a critical metric for NoC performance, as it directly impacts the response time of applications running on the MPSoC. For unicast packets, transmission delay is defined as the time taken from when the first flit of the packet leaves the source router to when the last flit arrives at the destination router. For multicast packets, we recorded the delay from the source router to each destination router separately and then calculated the average of these delays as the multicast packet's average transmission delay. This ensures that the metric accurately reflects the delay experienced by all destinations in the multicast group. We compared UBMR with the other three schemes under different network sizes (8×8 and 12×12 mesh), with packets containing varying numbers of unicast and multicast addresses (4, 8, 12 addresses per multicast packet). Figs. 9 and 10 show the average packet delay and network throughput under different conditions, respectively.

The results show the situation for different injection rates. Higher injection rates will lead to more contention, thereby increasing the delay.

In scenarios with small network sizes (8×8 mesh), fewer multicast addresses (4 addresses per packet), and low injection rates (e.g., 0.1 to 0.12 packets per cycle), such as shown in Fig. 9(a), the packet delay of the four multicast mechanisms is initially similar. This is because low traffic means there is little contention for network resources (buffers and crossbar), so even less optimized schemes like Banyan can perform well. The performance difference is minimal at a low injection rate of 0.1. Specifically, the average delay of UBMR is 16 clock cycles, while the delays for Conv, TLAM, and Banyan are 12, 14, and 13 clock cycles, respectively, indicating a negligible gap of 1–3 cycles. However, as network size increases (to 12×12 mesh and 16×16 mesh), injection rate increases (to 0.1 to 0.2 packets per cycle), and the number of multicast addresses increases (to 8, 12 or 16 addresses per packet), the gap in delay between the schemes gradually appears. This is because higher traffic leads to more contention, and the weaknesses of each scheme become more pronounced.

Notably, the Banyan scheme—due to the absence of buffers—experiences a significant increase in packet delay as injection rate rises. In Fig. 9(h) (12×12 mesh, 12–16 multicast addresses, injection rate 0.14), Banyan's average delay is 325 clock cycles, which is almost twice that of UBMR (185 cycles). This is attributed to the fact that in the absence of buffers, packets transmitted over the Banyan network are prone to deflection upon the occurrence of contention, where each deflection introduces additional hops to the packet transmission path and thus leads to an increase in transmission delay. For example, a packet destined for the east may be deflected west, then north, before finally reaching the destination, adding 5 or more hops to the original path.

TLAM adds an additional network layer on top of the traditional NoC (Conv), which allows it to handle more multicast traffic without excessive contention. This results in higher throughput than Conv (as shown in Fig. 10(a)), but TLAM uses a path-based multicast scheme, which requires packets to travel along a single path covering all destinations. This leads to a smaller reduction in average delay compared to UBMR. For example, in Fig. 9(d) (8×8 mesh, 8–12 multicast addresses, injection rate 0.2), TLAM achieves an average delay of 220 cycles, in contrast to UBMR which attains 189 cycles, representing a 14.1% reduction in latency. The longer delay of TLAM is due to the path-based scheme's longer hop count, especially when destinations are scattered.

The proposed UBMR scheme, due to buffer merging, experiences some performance degradation compared to Conv and TLAM in very small networks and low traffic. This is because merging buffers reduces the total buffer capacity, leading to slightly more contention in low traffic scenarios where individual buffers would otherwise be idle. Although the path-based multicast mechanism (used by TLAM) was partially adopted to compensate for this, at a smaller network size (8×8 mesh, 4 multicast addresses, injection rate 0.16), the average packet delay of UBMR decreases by 46.3% compared to Conv and 14.1% compared to Banyan. However, compared to TLAM, UBMR still reduces average packet delay by 24.6% in this scenario, highlighting its advantage over bufferless designs. In scenarios with large network sizes (16×16 mesh) and more multicast addresses (12–16 addresses per packet), as shown in Fig. 9(i), the advantages of UBMR become more pronounced. As expected, the Banyan scheme experiences a sharp increase in latency under high traffic loads. This is an inherent characteristic of bufferless designs, where frequent deflections occur to resolve contention. Unlike Banyan, UBMR retains a minimal set of unified buffers, which effectively prevents this latency saturation while still reducing buffer usage compared to traditional routers. Conv uses a unicast-based mechanism for multicast transmission, resulting in a large number of duplicate packets (as each multicast packet is split into multiple unicast packets). These duplicate packets rapidly deplete network bandwidth, thereby inducing network congestion and

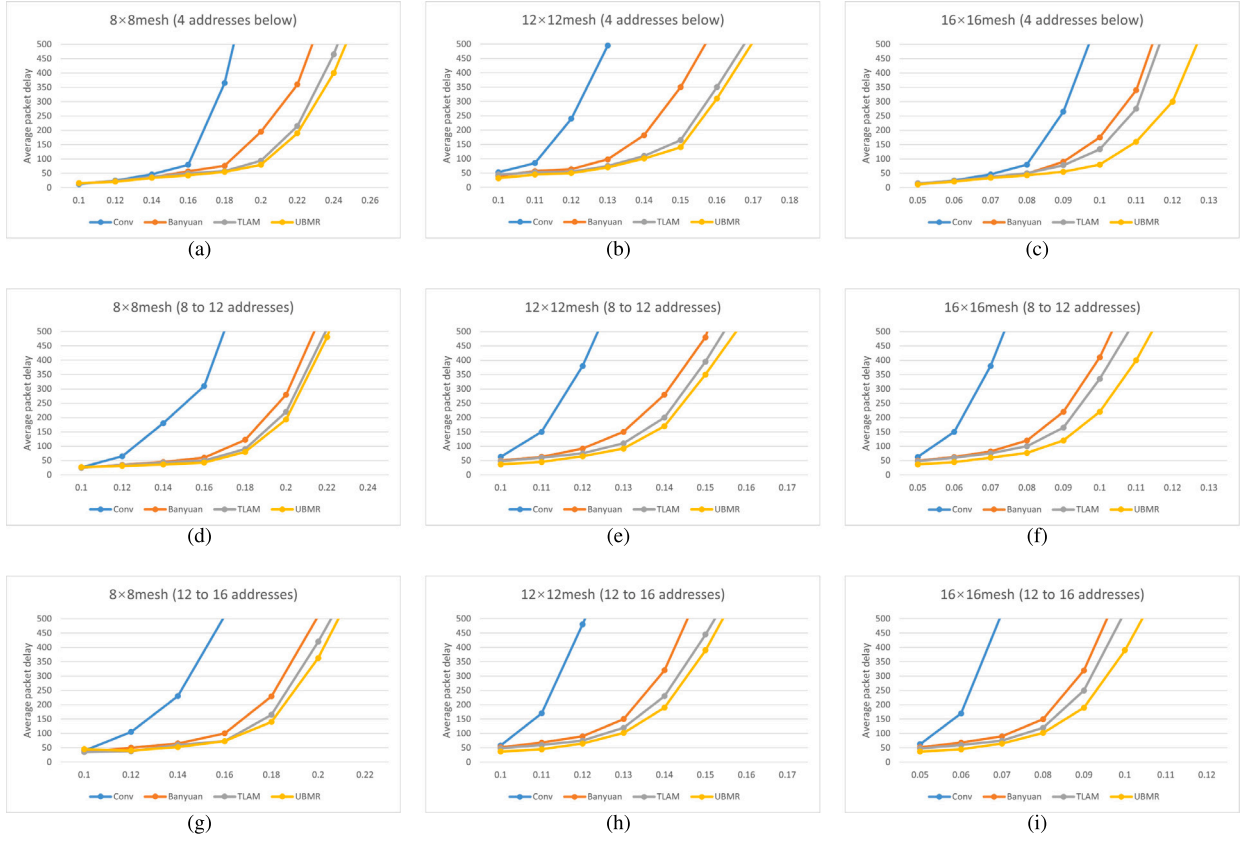


Fig. 9. Average packet delay under different network sizes and injection rates.

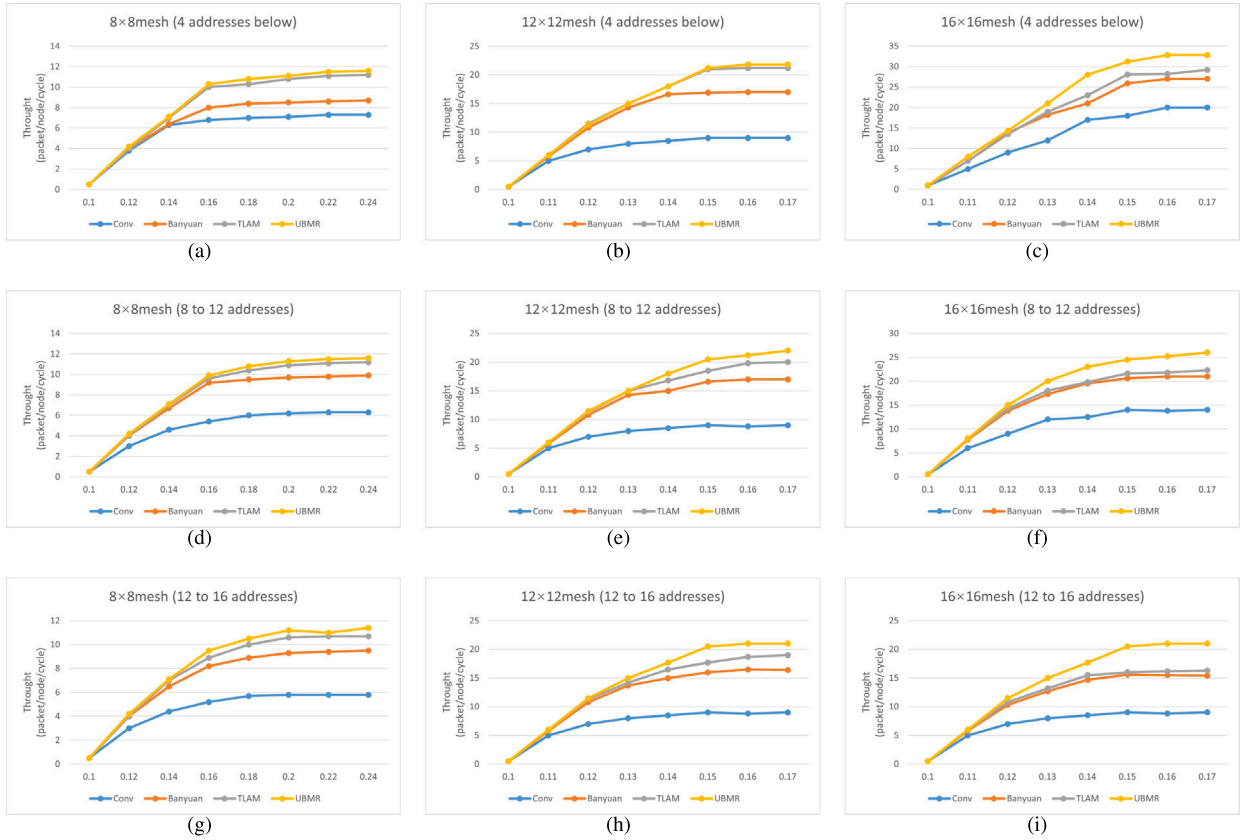


Fig. 10. Throughput under different network sizes and injection rates.

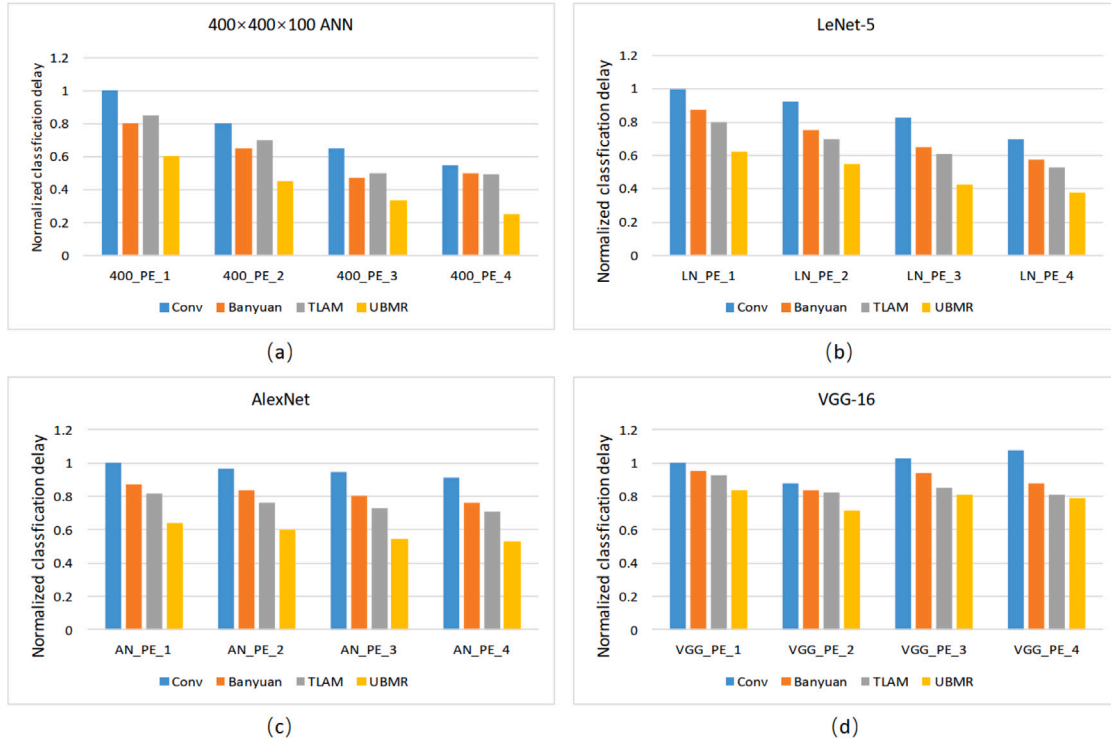


Fig. 11. Normalized classification delay.

an increase in transmission delay. TLAM uses a path-based transmission mechanism with a single transmission path. When there are more packets in the network (high injection rate), the critical path (the longest segment of the path) becomes easily congested, as all multicast packets in the same subnetwork share the same path. As the network injection rate in 0.08, the average packet delay of UBMR decreases by 32.1% compared to Banyuan and 15.2% compared to TLAM. This decrease is due to UBMR's efficient buffer utilization and tree-based multicast scheme, which reduces the number of packets in the network and minimizes contention. Under network size (16×16 mesh, 12 to 16 multicast addresses, injection rate 0.17), UBMR increases throughput by 28.8% compared to TLAM and 36.3% compared to Banyuan, as shown in Fig. 10(i). This demonstrates that UBMR maintains low delay even under high traffic loads, making it suitable for large scale NoCs.

5.3. Classification delay

Classification delay refers to the time from when the accelerator first reads parameters to when the final result is generated after mapping the deep neural network onto the NoC. We used UBMR and the other three schemes as multicast schemes for NoC-based DNN accelerators, with results shown in Fig. 11. The horizontal axis indicates the gradual increase in the number of neurons mapped in the Processing Element (PE), and the vertical axis indicates normalized classification delay.

Using the traditional unicast-based mechanism to process multicast packets results in a large number of duplicate packets in the accelerator, which rapidly consume network bandwidth and increase classification delay. Due to the relative clustering of multicast packet destination addresses in accelerators, TLAM's advantages (two tier network and bypass paths) cannot be fully utilized. Most accelerators generate bursty traffic, highlighting the disadvantage of Banyuan's bufferless design. For small DNN networks, Banyuan can still achieve relatively good performance, but as network size increases, classification delay rises significantly. UBMR reduces the average classification delay by 32% compared to Conv, and by 21% and 17% compared to Banyuan and TLAM, respectively.

5.4. Number of packets transmitted

The number of packets transmitted in the network is a key indicator of bandwidth efficiency. Unicast packets can be efficiently transmitted on existing NoC platforms, as they have only one destination and do not require replication. However, for multicast data packets, the adoption of unicast-based transmission (as employed in Conv) results in a massive flood of duplicate packets across the network, since each destination within the multicast group is delivered an individual unicast packet. These duplicates quickly deplete network bandwidth, leading to congestion and increased delay for all traffic (both unicast and multicast). We believe that an excellent multicast mechanism should effectively reduce the occurrence of redundant identical packets in the network, thereby saving bandwidth and improving overall network performance.

We compared the number of packets transmitted in the network when using UBMR and the other three schemes as multicast mechanisms for NoC-based DNN accelerators. The results are shown in Fig. 12, which presents the total number of packets transmitted during the classification of a single input image for different DNN models ($400 \times 400 \times 100$ ANN, LeNet-5, AlexNet, VGG-16) and PE counts (1 to 4). The number of packets is a critical metric for DNN accelerators because each transmission consumes power and bandwidth. Consequently, reducing the packet count directly leads to lower power consumption and less network congestion.

The traditional scheme (Conv) uses unicast to transmit multicast packets, so the number of packets transmitted is approximately equal to the number of multicast destinations multiplied by the number of multicast packets. For instance, in VGG-16 (a network with a substantial number of neurons), Conv transmits more than 1.2×10^9 packets when deployed on 4 PEs. This is due to the fact that each weight update necessitates the transmission of a multicast packet to hundreds of PEs, and each multicast packet is further split into hundreds of individual unicast packets for delivery. The number of redundant packets in the network is still very impressive, as shown in Fig. 12(d) (VGG-16, 4 PEs), where Conv's packet count is the highest among all schemes.

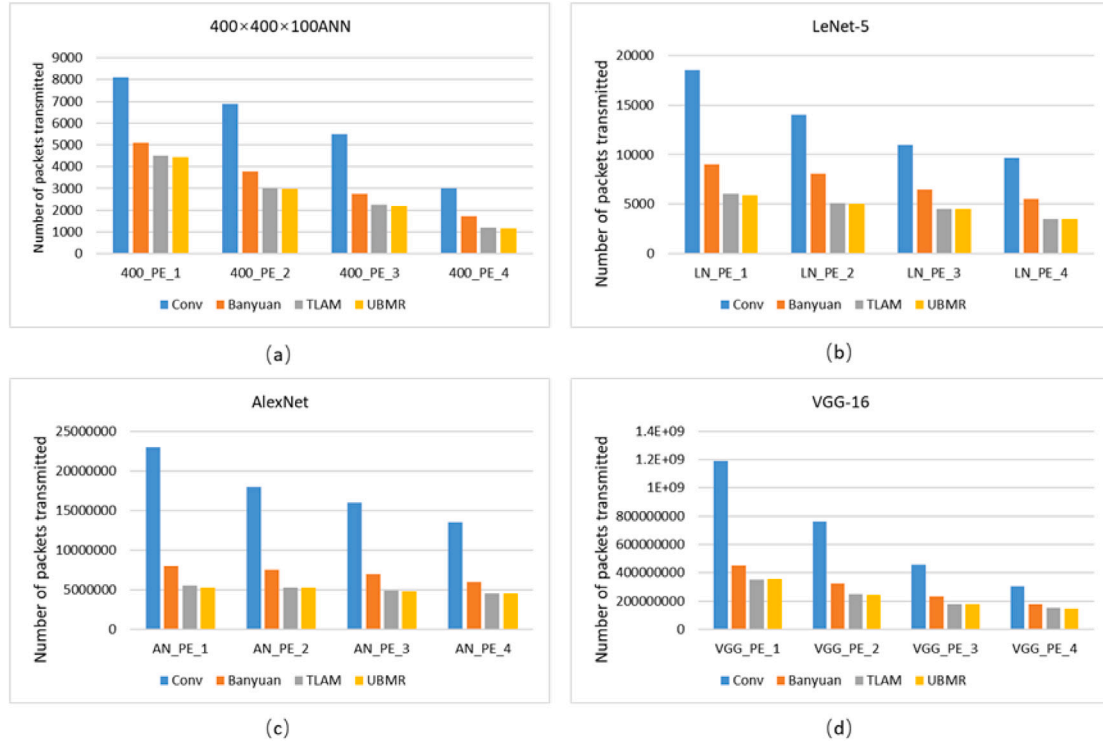


Fig. 12. Number of packets transmitted.

The Banyan scheme's multicast routing algorithm significantly reduces the number of packets in the network compared to Conv. Banyan replicates packets at the crossbar instead of splitting them into unicast packets, so the number of packets is approximately equal to the number of multicast packets multiplied by the average number of replications per packet. However, Banyan incurs a relatively higher frequency of packet replication in comparison to TLAM and UBMR. This is because Banyan lacks buffer resources and thus initiates packet replication at an earlier stage to mitigate deflection, which consequently results in a greater number of packet copies circulating in the network.

TLAM adopts a path-based multicast mechanism integrated with network partitioning and bypass path design. Given the inherent centralization of destination addresses for multicast packets in DNN accelerators, processing elements (PEs) are typically clustered to minimize transmission latency, and TLAM restricts packet replication exclusively to the bypass segments of the two-layer network architecture. This targeted replication strategy ensures that the majority of multicast packets avoid redundant duplication events, thereby contributing to a significantly reduced volume of packets circulating within the network.

For UBMR, the centralization of multicast packet destination addresses in DNN accelerators (PE clusters) means little or no packet replication is required during partitioning. When a multicast packet is split into subpackets, each subpacket's destinations are within a single PE cluster, so the initial branch router is often within the cluster, minimizing the need for replication. Additionally, within each partition, the mapping of neurons to NoCs results in the concentration of all destination addresses along either the X or Y dimension. For instance, PEs in a single cluster are mapped to routers with consecutive X coordinates. This means that multicast packets are rarely replicated, as there are no destinations in the opposite direction (Y direction for X concentrated destinations).

5.5. Power consumption and area

Fig. 13(a) and (b) show the router area and power consumption of the four schemes, respectively. The bufferless Banyan scheme achieves

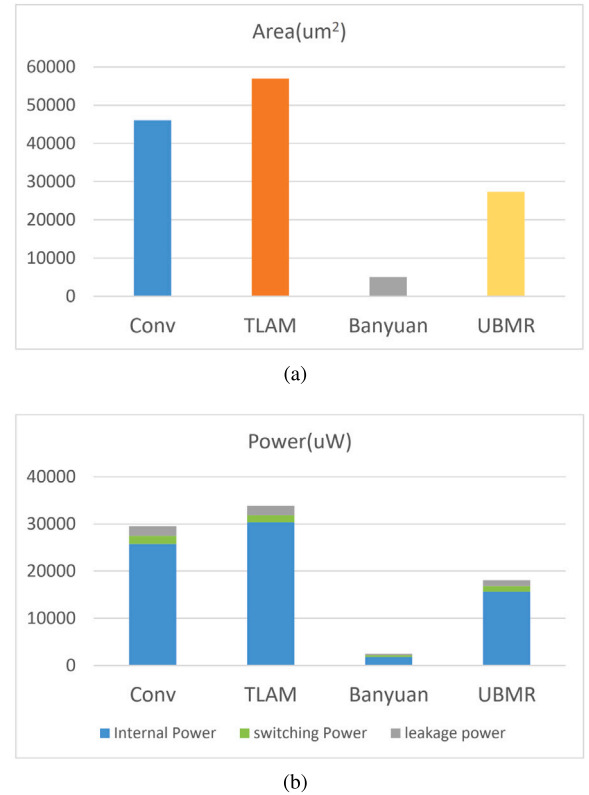


Fig. 13. Power consumption and area.

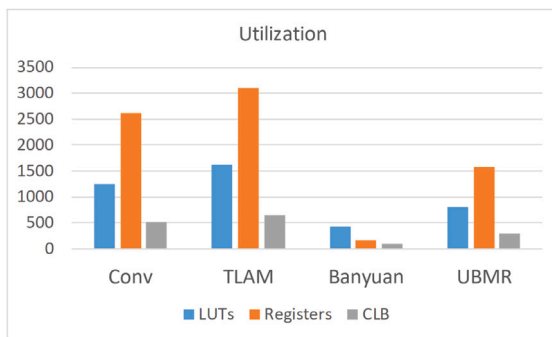


Fig. 14. Resource utilization in FPGA.

the lowest area and power consumption due to the complete elimination of input buffers. However, this comes at the cost of significant performance degradation at high injection rates. UBMR offers a balanced tradeoff: it consumes slightly more power than Banyan but delivers stable performance comparable to buffered routers (Conv and TLAM). Consequently, UBMR bridges the gap between fully buffered and bufferless architectures, achieving significant power savings without compromising network reliability. Compared to the buffered architectures (Conv and TLAM), UBMR significantly reduces area overhead by 40.6% and 52%, and power consumption by 38.6% and 47.1%, respectively, while ensuring lower average packet delay and higher network throughput. While Banyan is more power-efficient, it fails to meet performance requirements at high loads, making UBMR the more practical low-power solution.

5.6. Resource usage

Fig. 14 shows the resource usage (LUTs, Registers, and CLBs) of the four multicast mechanisms in FPGAs. The bufferless Banyan scheme uses the fewest resources but exhibits poor average packet delay performance at high injection rates and large network sizes. Due to buffer merging and crossbar simplification, UBMR reduces LUTs, Registers, and CLBs by 35%, 39.9%, and 43.6% compared to Conv, and by 50.4%, 49%, and 54.7% compared to TLAM, respectively.

6. Conclusion

In this paper, we propose a unified buffer multicast router architecture. By merging east–west and north–south direction buffers, the architecture reduces router area and power consumption. Additionally, to mitigate the increase in multicast packet delay caused by buffer merging, we introduce a tree-based multicast mechanism. This mechanism splits multicast packets through NoC network partitioning and uses a multicast address labeling strategy to prevent infinite packet replication. The proposed mechanism is also deadlock-free. Detailed simulation results demonstrate that UBMR achieves an optimal balance between performance and cost. Compared to traditional buffered routers (Conv and TLAM), it significantly reduces area and power consumption. Meanwhile, compared to bufferless designs (Banyan), UBMR maintains low latency and high throughput under heavy traffic loads, proving its suitability for high performance NoCs. Detailed simulation results demonstrate that compared to Conv and TLAM, UBMR reduces area overhead by 40.6% and 52%, and power consumption by 38.6% and 47.1%, respectively, with only an 8.6% and 23.2% increase in average packet delay. Although buffer merging may cause slight performance loss, the benefits in terms of area and power savings are significantly more substantial.

CRediT authorship contribution statement

Chenglong Sun: Writing – original draft, Methodology, Software, Funding acquisition. **Jiaxin Wang:** Formal analysis, Conceptualization. **Yiming Ouyang:** Validation, Resources. **Qi Wang:** Supervision, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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